

Multiple Choice (10 marks)

1. Which of the following confirmed values meet the diagnostic threshold for diabetes?
 - (a) fasting blood glucose ≥ 160 mg/dl
 - (b) fasting blood glucose ≥ 140 mg/dl
 - (c) fasting blood glucose ≥ 126 mg/dl
 - (d) random glucose > 180 mg/dl
 - (e) fasting blood glucose ≥ 120 mg/dl
2. Simple tandem repeat polymorphisms in humans are most useful for
 - (a) studying the effects of radiation exposure
 - (b) identifying the presence of bacterial infections
 - (c) solving criminal and paternity cases
 - (d) determining blood type
 - (e) transferring disease resistance factors into bone marrow cells
3. Proprioceptive nerve endings in synovial joints are located in
 - (a) synovial fluid and articular discs.
 - (b) capsule and ligaments.
 - (c) articular cartilage and ligaments.
 - (d) capsule and articular cartilage.
 - (e) synovial membrane and capsule.
4. Severe acute malnutrition in young children is defined as:
 - (a) Weight-for-age Z score < -2 and oedema
 - (b) Height-for-age Z score < -2 or weight-for-height Z score < -2 and oedema
 - (c) Height-for-age Z score < -3 and oedema
 - (d) Weight-for-age Z score < -2 or height-for-age Z score < -2 or oedema
 - (e) Height-for-age Z score < -3 or weight-for-age Z score < -3 and oedema
5. What is the difference between a male and a female catheter?
 - (a) Female catheters are used more frequently than male catheters.
 - (b) Male catheters are bigger than female catheters.
 - (c) Male catheters are more flexible than female catheters.
 - (d) Male catheters are made from a different material than female catheters.
 - (e) Female catheters are longer than male catheters.

True / False (10 marks)

Answer True or False

6. True or False: Grade 2/5 power represents movement of the arm only against gravity.
7. True or False: A higher level of education is linked to an increased risk of Alzheimer's disease in older adults.
8. True or False: The temporal bone has a process that articulates with the zygomatic bone to form the zygomatic arch.
9. True or False: Type I muscle fibers are characterized as red, oxidative, and slow contracting, primarily used for endurance activities.
10. True or False: The concentration of a substance in food is a critical factor in assessing its safety for human consumption.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the genetic basis of DiGeorge/Shprintzen syndrome, focusing on the implications of the chromosomal deletion involved and its effects on health and development.
12. Discuss the role of ubiquitin in proteolysis, specifically addressing its attachment to proteins designated for synthesis versus catabolism and the implications for cellular regulation.

End of Paper